

ZYRA Protocol: A Decentralized Agentic AI Swarm Network

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Abstract

A purely peer-to-peer version of artificial intelligence execution would allow complex, autonomous agentic workflows to be computed directly between unassociated nodes without relying on centralized orchestration servers. We propose a decentralized protocol, ZYRA, which establishes a trustless ecosystem for the execution of Large Language Models (LLMs) and autonomous agents. The network introduces a novel consensus algorithm, Proof-of-Useful-Work (PoUW), wherein idle computational resources are utilized to solve real-world programmatic tasks. Security and validation are enforced via a dynamic P2P Gossip Network, where randomly selected Smart Judges evaluate execution trajectories locally. To mitigate state bloat on the underlying Celo blockchain, the protocol leverages InterPlanetary File System (IPFS) integration via Content Identifiers (CIDs).

1. Introduction

The proliferation of Large Language Models has historically been restricted by centralized hardware monopolies and prohibitive computational costs. While distributed computing frameworks exist, they inherently suffer from the Byzantine Generals Problem in evaluating the stochastic nature of AI outputs. In this paper, we propose a novel cryptographic framework, the ZYRA Protocol, designed to coordinate autonomous AI swarms over a decentralized topology.

2. Architecture: P2P Gossip Network

In its initial iteration, the ZYRA protocol relied upon a centralized tracker to adjudicate computational validity. To achieve full decentralization, Version 2.0 transitions to a Peer-to-Peer (P2P) Gossip architecture. Let N be the set of active

nodes in the network. When a node $N_i \in N$ successfully executes a task, it generates an execution trajectory T_i .

Instead of submitting T_i to a central arbiter, N_i broadcasts T_i to its peer cohort. A stochastic selection algorithm assigns a subset of peers $J \subseteq N$ to act as *Smart Judges*. Each judge $N_j \in J$ independently verifies the logical coherence and execution results of T_i using localized, quantized LLM inference models (e.g., Q4_K_M representation). If a quorum $q > \frac{1}{2}$ is reached, consensus is achieved.

3. Proof-of-Useful-Work (PoUW) Consensus

Traditional Proof-of-Work (PoW) mechanisms secure networks through energy-intensive mathematical inversions of cryptographic hashes, such as SHA-256. ZYRA replaces this arbitrary computation with Proof-of-Useful-Work (PoUW). The computational expenditure is directly applied to processing stochastic inference pipelines. The probability of a node mining the next block is proportional to the computational complexity and utility of the task resolved.

4. IPFS State Compression

A significant bottleneck in bridging stochastic AI output with deterministic smart contracts is payload dimensionality. An execution trajectory T_i often contains extensive Python source code, abstract syntax trees (ASTs), and verbose terminal standard outputs.

To achieve $O(1)$ on-chain storage complexity, ZYRA integrates the InterPlanetary File System (IPFS). The raw payload T_i is compressed and pinned to the IPFS network, which yields a deterministic Content Identifier (CID_i). The smart contract C deployed on the Celo network strictly records CID_i . Thus, the on-chain payload size remains fixed at approximately 46 bytes, dramatically reducing transaction latency and gas expenditure.

5. Tokenomics and Economic Incentives

The network is sustained by a native cryptographic asset, \$ZYRA. The monetary policy strictly enforces a hard cap of 21,000,000 units. The reward function $R(t)$ is dynamically adjusted based on network difficulty and mempool saturation. Miners are incentivized to maintain high-availability hardware to maximize probability $P(\text{mint})$, whereas bad actors submitting fabricated execution trajectories are systematically penalized through stake slashing mechanisms.

6. Conclusion

We have proposed a system for decentralized AI orchestration that does not rely on trust. By coupling a robust P2P Gossip validation layer with IPFS state

compression and a PoUW consensus model, the ZYRA protocol provides a scalable, censorship-resistant infrastructure for the future of agentic machine learning.

References:

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- [2] Buterin, V. (2013). Ethereum Whitepaper.
- [3] Benet, J. (2014). IPFS - Content Addressed, Versioned, P2P File System.
- [4] The ZYRA Foundation. (2026). ZYRA Github Repository and Architecture Specifications.